

## 75/25 YES/NO Stress Test Results on POPE Benchmark

Ground truth label ratio: 75% YES / 25% NO

Baseline is the reference point for FN→TP and TN→FP columns.

### Column definitions:

- **FN→TP**: Change in correctly answered YES questions relative to baseline (positive = more correct YES answers recovered).
- **TN→FP**: Change in incorrectly answered YES questions relative to baseline (positive = more wrong YES answers introduced).
- **Yes%**: Proportion of questions answered “yes” by the model.
- **Acc%**: Overall accuracy on the 75/25 filtered benchmark.

**Note:** Each image retains 3 YES and 1 NO question (2000 total per split). The same NO question ordinals are deleted consistently across all POPE variants (Random, Popular, Adversarial) and all decoding methods for each image (seed = 42). **CD methods:** VCD, SID, ICD. **Spurious baselines:** PBA (prompt: “Answer yes whenever possible.”), OLM (output label manipulation).

Table 1: 75/25 POPE stress test – **LLaVA-v1.5-7B** on **GQA**.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 66.1        | 87.50 |
|             | VCD      | +67                 | +51   | 72.0        | 88.30 |
|             | SID      | +51                 | +31   | 70.2        | 88.50 |
|             | ICD      | −30                 | −5    | 64.3        | 86.25 |
|             | PBA      | +46                 | +16   | 69.2        | 89.00 |
|             | OLM      | +113                | +49   | 74.2        | 90.70 |
| Popular     | Baseline | ref                 | ref   | 68.0        | 85.55 |
|             | VCD      | +70                 | +48   | 74.0        | 86.65 |
|             | SID      | +50                 | +21   | 71.6        | 87.00 |
|             | ICD      | −29                 | −10   | 66.1        | 84.60 |
|             | PBA      | +46                 | +32   | 72.0        | 86.25 |
|             | OLM      | +113                | +92   | 78.3        | 86.60 |
| Adversarial | Baseline | ref                 | ref   | 69.9        | 83.70 |
|             | VCD      | +74                 | +39   | 75.5        | 85.45 |
|             | SID      | +46                 | +32   | 73.8        | 84.40 |
|             | ICD      | −30                 | −4    | 68.2        | 82.40 |
|             | PBA      | +46                 | +14   | 72.9        | 85.30 |
|             | OLM      | +113                | +86   | 79.8        | 85.05 |

Table 2: 75/25 POPE stress test – **LLaVA-v1.5-7B** on **COCO**. ICD outputs not available.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 58.2        | 81.85 |
|             | VCD      | +96                 | +18   | 63.9        | 85.75 |
|             | SID      | +69                 | +12   | 62.3        | 84.70 |
|             | PBA      | +37                 | +1    | 60.2        | 83.65 |
|             | OLM      | +126                | +17   | 65.4        | 87.30 |
| Popular     | Baseline | ref                 | ref   | 58.9        | 81.20 |
|             | VCD      | +103                | +27   | 65.4        | 85.00 |
|             | SID      | +67                 | +24   | 63.4        | 83.35 |
|             | PBA      | +37                 | +6    | 61.1        | 82.75 |
|             | OLM      | +126                | +26   | 66.5        | 86.20 |
| Adversarial | Baseline | ref                 | ref   | 59.5        | 80.50 |
|             | VCD      | +103                | +49   | 67.1        | 83.20 |
|             | SID      | +65                 | +34   | 64.5        | 82.05 |
|             | PBA      | +36                 | +8    | 61.7        | 81.90 |
|             | OLM      | +126                | +50   | 68.3        | 84.30 |

Table 3: 75/25 POPE stress test – **LLaVA-v1.5-7B** on **AOKVQA**.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 64.8        | 86.75 |
|             | VCD      | +69                 | +34   | 69.9        | 88.50 |
|             | SID      | +44                 | +22   | 68.0        | 87.85 |
|             | ICD      | −26                 | −2    | 63.3        | 85.55 |
|             | PBA      | +42                 | +6    | 67.2        | 88.55 |
|             | OLM      | +105                | +40   | 72.0        | 90.00 |
| Popular     | Baseline | ref                 | ref   | 66.6        | 84.90 |
|             | VCD      | +63                 | +45   | 72.0        | 85.80 |
|             | SID      | +34                 | +22   | 69.4        | 85.50 |
|             | ICD      | −30                 | −4    | 64.9        | 83.60 |
|             | PBA      | +42                 | +16   | 69.5        | 86.20 |
|             | OLM      | +105                | +67   | 75.2        | 86.80 |
| Adversarial | Baseline | ref                 | ref   | 69.8        | 81.65 |
|             | VCD      | +65                 | +55   | 75.8        | 82.15 |
|             | SID      | +41                 | +25   | 73.2        | 82.45 |
|             | ICD      | −30                 | −12   | 67.8        | 80.75 |
|             | PBA      | +42                 | +21   | 73.0        | 82.70 |
|             | OLM      | +105                | +79   | 79.0        | 82.95 |

Table 4: 75/25 POPE stress test – **Qwen2.5-VL-7B** on **GQA**.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 65.0        | 88.30 |
|             | VCD      | +3                  | +2    | 65.2        | 88.35 |
|             | SID      | +20                 | +5    | 66.2        | 89.05 |
|             | ICD      | −5                  | +1    | 64.8        | 88.00 |
|             | PBA      | +9                  | −1    | 65.4        | 88.80 |
|             | OLM      | +71                 | +13   | 69.2        | 91.20 |
| Popular     | Baseline | ref                 | ref   | 66.5        | 87.05 |
|             | VCD      | −3                  | +6    | 66.7        | 86.60 |
|             | SID      | +17                 | +6    | 67.7        | 87.60 |
|             | ICD      | −15                 | −3    | 65.6        | 86.45 |
|             | PBA      | +6                  | +3    | 67.0        | 87.20 |
|             | OLM      | +68                 | +17   | 70.8        | 89.60 |
| Adversarial | Baseline | ref                 | ref   | 68.4        | 85.20 |
|             | VCD      | −4                  | +3    | 68.3        | 84.85 |
|             | SID      | +17                 | +6    | 69.5        | 85.75 |
|             | ICD      | −10                 | +1    | 68.0        | 84.65 |
|             | PBA      | +6                  | +2    | 68.8        | 85.40 |
|             | OLM      | +68                 | +23   | 73.0        | 87.45 |

Table 5: 75/25 POPE stress test – **Qwen2.5-VL-7B** on **COCO**.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 58.9        | 83.35 |
|             | VCD      | +16                 | 0     | 59.7        | 84.15 |
|             | SID      | +17                 | +1    | 59.8        | 84.15 |
|             | ICD      | −11                 | −1    | 58.2        | 82.85 |
|             | PBA      | +13                 | 0     | 59.5        | 84.00 |
|             | OLM      | +72                 | +3    | 62.6        | 86.80 |
| Popular     | Baseline | ref                 | ref   | 59.2        | 82.95 |
|             | VCD      | +18                 | 0     | 60.2        | 83.85 |
|             | SID      | +17                 | +4    | 60.3        | 83.60 |
|             | ICD      | −8                  | 0     | 58.9        | 82.55 |
|             | PBA      | +13                 | −1    | 59.9        | 83.65 |
|             | OLM      | +72                 | +7    | 63.2        | 86.20 |
| Adversarial | Baseline | ref                 | ref   | 59.8        | 82.75 |
|             | VCD      | +14                 | +4    | 60.7        | 83.25 |
|             | SID      | +14                 | +11   | 61.0        | 82.90 |
|             | ICD      | −10                 | +1    | 59.3        | 82.20 |
|             | PBA      | +13                 | +1    | 60.5        | 83.35 |
|             | OLM      | +72                 | +15   | 64.0        | 85.50 |

Table 6: 75/25 POPE stress test – **Qwen2.5-VL-7B** on **AOKVQA**.

| Split       | Method   | YES Response Change |       | Performance |       |
|-------------|----------|---------------------|-------|-------------|-------|
|             |          | FN→TP               | TN→FP | Yes%        | Acc%  |
| Random      | Baseline | ref                 | ref   | 65.5        | 88.60 |
|             | VCD      | +11                 | +3    | 66.2        | 89.00 |
|             | SID      | +21                 | +1    | 66.6        | 89.60 |
|             | ICD      | −18                 | −1    | 64.5        | 87.75 |
|             | PBA      | +7                  | −1    | 65.8        | 89.00 |
|             | OLM      | +60                 | +9    | 69.0        | 91.15 |
| Popular     | Baseline | ref                 | ref   | 66.6        | 87.45 |
|             | VCD      | +4                  | +3    | 67.0        | 87.50 |
|             | SID      | +21                 | +5    | 68.0        | 88.25 |
|             | ICD      | −20                 | −4    | 65.5        | 86.65 |
|             | PBA      | +7                  | +2    | 67.1        | 87.70 |
|             | OLM      | +60                 | +12   | 70.2        | 89.85 |
| Adversarial | Baseline | ref                 | ref   | 69.2        | 84.85 |
|             | VCD      | +1                  | −1    | 69.2        | 84.95 |
|             | SID      | +21                 | +3    | 70.5        | 85.75 |
|             | ICD      | −20                 | +3    | 68.4        | 83.70 |
|             | PBA      | +7                  | +5    | 69.8        | 84.95 |
|             | OLM      | +60                 | +19   | 73.2        | 86.90 |